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Expanding the Therapeutic Reach of Chimeric Antigen Receptor-T Cells and Bispecific T Cell Engagers Across Solid Tumors

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Abstract

The introduction of T cell-based therapeutics in hematologic malignancies has led to improvements in outcomes for patients with acute leukemia, lymphoma, and multiple myeloma. To date, the Food & Drug Administration (FDA) has approved 7 chimeric antigen receptor-T (CAR-T) cell therapies and 7 bispecific T cell engagers (BiTEs) across a variety of hematologic malignancies; however, the extension of CAR-T therapies and BiTEs to the solid tumor arena has been somewhat limited. In this review, we discuss the landmark data that led to the commercialization of 4 novel FDA-approved T cell-based therapeutics in solid malignancies, including tarlatamab for small cell lung cancer, afamitresgene autoleucel for synovial sarcoma, lifileucel for metastatic melanoma, and tebentafusp for metastatic uveal melanoma. We discuss the targetable antigen landscape of CAR-T therapies and BiTEs under investigation in solid malignancies. We explore the translational potential for various CARs under active investigation, including HER2-directed CARs in breast cancer, PSCA-directed CARs for prostate cancer, EGFR-IL13Ra2 and EGFR-vIII CARs for glioblastoma, and GD2-directed CARs for neuroblastoma. We glean from lessons learned for existing CAR-T therapies and BiTEs for hematologic malignancies and emphasize solutions toward facilitating the clinical rollout of T cell-based therapies in solid tumors, including scalability to meet the growing needs of clinical oncology. Some solutions include addressing on-target, off-tumor toxicity; improving the manufacturing of CARs; optimizing the tissue-specific tumor microenvironment by combating immune desert tumors; and discovering natural tumor neoantigens and non-self epitopes generated by tumor-specific mutations. These concepts can help provide transformative benefits for patients with solid malignancies in the coming years.

Keywords

CAR-T; BiTEs; tumor antigens; cell therapy

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I. INTRODUCTION

An understanding of effector T cell biology has been essential in developing promising immunological treatments for malignancies, including hematological cancers and solid tumors. Specifically, chimeric antigen receptor-T (CAR-T) technology has paved the way for delivery of genetically modified autologous T cells to patients with leukemia and lymphoma.^{1,2} Since their inception, these CAR-T therapies have undergone significant manufacturing improvement to become increasingly powerful and specialized in unique clinical settings. Of note, due to the circulating nature of leukemias and lymphomas, CAR-T therapies have been deployed almost exclusively in hematologic malignancies thus far; however, they offer a strong possible benefit in solid tumor cases if certain hurdles can be overcome. To date, the Food and Drug Administration (FDA) has approved CAR-T therapies across varying indications (such as acute lymphoblastic leukemia, lymphoma, and multiple myeloma), but the approval of CAR-T therapies in the solid tumor space is very limited.^{3–6}

Bispecific T cell engagers (BiTEs) join both T cells and tumor cells in order to draw an immune response to the site of the cancer.^{7,8} BiTEs offer the potential to initiate an immune attack on the tumor without intrinsic modification to the patient's own immune system and can thus be used as off-the-shelf therapy. BiTEs also have the added benefit of bypassing histocompatibility barriers. Due to the clear promise demonstrated by the entire class of T cell-mediated therapies, further examination of the safety and efficacy of each candidate therapeutic is essential for advancements in clinical oncology.

A remaining hurdle to the widespread use of CAR-T and BiTE therapies is that solid tumors are often resistant to these therapies.⁹ Some reasons underlying the limited efficacy may be related to immune desert ("cold") tumor microenvironment (TME) or limited ability of T cells to migrate into resistant tumors. This review discusses some challenges and solutions to bringing T cell-based therapies to patients with solid malignancies.

II. FDA APPROVALS FOR CAR-T THERAPIES AND BiTEs IN HEMATOLOGIC MALIGNANCIES

The current CAR-T therapies in hematologic malignancies include tisagenlecleucel, axicabtagene ciloleucel, lisocabtagene maraleucel, brexucabtagene autoleucel, idecabtagene vicleucel, ciltacabtagene autoleucel, and obecabtagene autoleucel (Table 1).^{5,6,10–13} (Roddie *et al.*, *NEJM* 2024) Adverse effects of CAR-T therapies include severe cytopenias, cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS), and immune effector cell-associated hemophagocytic lymphohistiocytosis-like syndrome.¹⁴

Tisagenlecleucel (tisa-cel) was the first fully approved CAR-T cell therapy for use in 2017 after the landmark ELIANA trial showed favorable results in relapsed and refractory B cell acute lymphocytic leukemia (B-ALL).³ Tisa-cel was then studied in the JULIET and ELARA trials, which eventually led to approval of tisa-cel as third-line therapy in large B-cell lymphoma and follicular lymphoma.^{4,15} Axicabtagene ciloleucel (axi-cel) was studied in the ZUMA-1 and ZUMA-5 trials and is now approved in second-line refractory or short-term relapsed large B-cell lymphoma as well as all third line follicular

lymphoma cases.^{5,16} Lisocabtagene maraleucel (liso-cel) was studied in the TRANSCEND NHL 001 and TRANSFORM trials. This cell therapy has shown activity against large B cell lymphoma, follicular lymphoma, mantle cell lymphoma, and chronic lymphocytic leukemia (CLL), and these seminal clinical trials have led to the approval of this cell therapy in the second-line and short-term relapse setting in all of these conditions.^{10,17} Brexucabtagene autoleucel (brexu-cel) was studied in the ZUMA-2 and ZUMA-3 trials.^{12,18} These trials led to its approval in the relapsed or refractory mantle cell lymphoma and B-ALL spaces. In patients with refractory multiple myeloma, idecabtagene vicleucel (ide-cel) and ciltacabtagene autoleucel (cilta-cel) were studied in trials by Munshi et al. and Berdeja et al., respectively, and both products exhibit BCMA-binding capacity.^{6,19} They were initially approved in the fifth-line setting, but now are approved in the third-line and second-line settings, respectively. Finally, obecabtagene autoleucel (obe-cel) was recently approved for relapsed/refractory B-ALL, based on the FELIX trial.¹³ This was a phase 1B-2 trial showing a median overall survival (OS) of 15.6 months. This CAR-T therapy showed durable responses with low incidence of grade 3 or higher immune-related adverse events.

In addition to these therapies, a total of 7 BiTEs have been approved by the FDA for hematologic diseases (Table 1). Blinatumomab, approved for use in relapsed and refractory B-ALL, was first studied in the TOWER trial. This BiTE operates by linking CD3 on T cells to CD19 on the tumor for an anti-cancer response.⁷ Epcoritamab, glofitamab, and mosunetuzumab are BiTEs with CD3 and CD20 linkage modalities, and each of these therapies was introduced commercially in 2022.^{20–22} Epcoritamab is approved for use in third-line and beyond for large B cell lymphoma and follicular lymphoma, while glofitamab is only approved for the former and mosunetuzumab for the latter.

Three BiTEs were recently approved for refractory multiple myeloma. Teclistamab and elranatamab are anti-CD3/anti-BCMA therapies that stem from the MajesTEC-1 and MagnetisMM-3 trials, respectively, and both are approved in the fifth-line and beyond.^{8,23} Finally, talquetamab was tested in the MonumenTAL-1 trial and also exists as an option in the fifth-line and beyond for patients.²² The unique aspect of talquetamab is that it has specificity for GPRC5D, an orphan receptor on myeloma cells²⁴.

III. FDA APPROVALS FOR CAR-T THERAPIES AND BiTEs IN SOLID MALIGNANCIES

III.1. Tarlatamab

The first FDA-approved BiTE in solid malignancies was tarlatamab for patients with previously treated small cell lung cancer (Figure 1) (Table 2).²⁵ Tarlatamab targets small cell lung cancer by binding both the CD3 receptor of the T cell and delta-like ligand 3 (DLL3) expressed in most small cell lung tumors. Small cell lung cancer is an especially difficult disease to treat as its intrinsic TME suppresses nearby immune cells from any anti-cancer potency by decreasing expression of MHC class I and other mechanisms. In order to circumvent this challenge, tarlatamab forgoes the MHC dependency of T cell-mediated attack. DLL3 is estimated to be expressed in about 90% of small cell lung cancer.

The Phase 2 DeLLphi-301 trial studied tarlatamab in 220 patients with relapsed/refractory small cell lung cancer.²⁵ The overall response rate (ORR) was 40% in the 10 mg dosing cohort and 32% in the 100 mg dosing cohort. The 9-month OS was 68% and 66%, respectively, for these 2 groups. Approximately 59% of patients had a 6-month durability of response. CRS was the most significant adverse effect, affecting 51–61% of treated patients. Based on this data, the FDA approved tarlatamab in May 2024.

III.2. Afamitresgene autoleucel (afami-cel)

Afami-cel is an HLA-restricted T cell therapy that is generated by *ex vivo* transduction of a patient's T cells so that they express an affinity-enhanced receptor for human melanoma antigen A4 (MAGE-A4).²⁶ MAGE-A4 is overexpressed in synovial sarcoma, as well as other tumors. MAGE-A4 is processed inside the malignant cell, allowing fragments of peptides to become evident on the cell's surface. These fragments can then be recognized by lower affinity receptors on the T cell, but the *ex vivo* transduction allows higher affinity receptors to increase recognition of the tumor. Afami-cel infiltrates tumors and triggers an IFN- γ response to exert its anti-tumor effects.

In a multi-center Phase 1 trial, afami-cel was studied in 38 patients with relapsed/refractory synovial sarcoma, head and neck cancer, or ovarian cancer. The ORR was 24%.²⁶ All patients unfortunately developed grade 3 or worse hematologic toxicity. The SPEARHEAD-1 trial then studied the effects of afami-cel in patients with synovial sarcoma and myxoid round cell liposarcoma.²⁷ With a trial size of 44 patients with synovial sarcoma, a durable response rate of 39% was observed. CRS and cytopenias unfortunately affected well over half of the patient population. The trial demonstrated such promise as to warrant the first FDA approval of a T cell therapy in the treatment of solid malignancies as of August 2024 (Figure 1). This FDA approval attests to the value of the clinical benefit of CAR-T therapy in this context, especially given that synovial sarcoma is a relatively rare tumor.

III.3. Lifileucel

In the 1990s, recombinant human IL-2 was studied in an effort to improve expansion of T cells such that these cells could target tumors. Tumor-infiltrating lymphocytes (TILs) were presumed to target melanoma expressing tumor-associated antigens, such as MART-1, gp100, and tyrosinase. These TILs mostly comprise polyclonal T cells that harbor diverse antigen specificity.²⁸ TILs can be expanded *ex vivo* after extraction from the TME and can attack multiple tumor-specific neoantigens.

In a Phase 2 single-arm, multi-center trial, lifileucel was studied in 66 patients with metastatic or unresectable melanoma.²⁸ Patients received non-myeloablative lymphodepleting chemotherapy followed by a one-time infusion of lifileucel and high-dose IL-2. The ORR was 36%. At a median follow-up of 18.7 months, median duration of response was not reached.²⁸ Lifileucel thereby became the first-in-class, first-in-human autologous TIL approved by the FDA.

TIL therapy has also been compared to the current standard of care. In a Phase 3 multi-center trial of patients with unresectable stage III or IV melanoma, patients were randomized 1:1 to receive either TILs or anti-CTLA-4 therapy (ipilimumab).²⁹ Median OS was 25.8

months vs. 18 months, respectively. Median progression-free survival (PFS) was 7.2 months vs. 3.1 months, respectively.²⁹

III.4. Tebentafusp

Tebentafusp is a BiTE comprised of an anti-CD3 effector conjoined with a soluble affinity-enhanced T cell receptor, which directs T cells to gp100-positive melanoma cells. This BiTE was studied in the IMCgp100–202 trial, a Phase 3 trial of patients with metastatic uveal melanoma.³⁰ The 1-year OS was 73% and PFS was 31%; these metrics were superior for the tebentafusp group compared to the investigator's choice chemotherapy group. Long-term follow-up results confirmed OS benefit, with median OS of 21.6 months.³¹

IV. INVESTIGATIONAL T CELL-BASED THERAPIES IN SOLID MALIGNANCIES

IV.1. HER2-directed CAR-T cells

HER2 is a well-established therapeutic target for various cancers, including breast, gastric, biliary tract, colorectal, non-small cell lung, and bladder cancers.³² *In vitro* and xenograft models have shown that HER2-specific CAR-T cells can result in sustained antitumor effect against HER2-positive tumors.^{33–36} The success of current HER2-targeting drugs has spurred interest in developing HER2-specific CAR-T cell therapies. One notable advantage of using CAR-T cells to target HER2 is their potential effectiveness in cancers where HER2 is overexpressed but not gene-amplified, a scenario in which monoclonal antibodies often fall short.^{37–38}

In clinical studies, a Phase 1/2 study explored the efficacy of HER2-specific CAR-T cells in treating sarcoma.³⁹ Among the 19 enrolled patients, the median OS was 10.3 months. Of the 17 evaluable participants, 4 had stable disease. Another Phase 1 clinical trial assessed the feasibility of HER2-specific CAR-T cells in patients with biliary tract and pancreatic cancers.⁴⁰ In this study, 11 patients received a median dose of 2.1×10^6 CAR(+) cells per kg, with 1 participant achieving a partial response and 5 achieving stable disease. Of note, 1 case of severe upper gastrointestinal hemorrhage occurred. The median PFS was 4.8 months. Additionally, Hegde et al. reported results from a prospective Phase 1 trial involving HER2-specific CAR-T cells for advanced sarcoma.⁴¹ Thirteen participants were treated, and the dose-limiting toxicity was CRS. The ORR was 50%. The median OS for all participants was 8.2 months.

IV.2. PSCA-directed CAR-T cells

A variety of tumor-associated antigens are expressed in prostate cancer, including prostate-specific antigen (PSA), prostate stem cell antigen (PSCA), and prostate acid phosphatase (PAP). CARs have been generated against these antigens for patients with advanced prostate cancer, including castrate-resistant prostate cancer.⁴² The leading target antigen has been PSCA.

In a Phase 1 trial involving 14 patients with metastatic castrate-resistant prostate cancer, PSCA-directed CAR-T cells were studied.⁴³ Across the 3 dose levels of the study, 4 of the

14 participants experienced a decline in prostate specific antigen (PSA) levels, indicating some degree of biochemical response. In most of the significant responses, the positive effect was unfortunately not maintained for longer than the first month following treatment. Across all dose levels, 5 of the 14 patients suffered from low to moderate grade CRS, while no other major toxicities were reported across multiple patients. Overall, this CAR-T cell treatment offered a moderate but transient benefit to the subset of patients without major toxicity issues.

IV.3. EGFR-IL13R α 2 CAR-T cells

Glioblastoma is an aggressive central nervous system malignancy that is characterized by poor survival rates and limited targeted therapeutic options, with 5-year survival rate of approximately 5%.⁴⁴ A subset of tumor cells refractory to treatment are thought to initiate recurrence in most patients. CAR-T cell therapy targeting IL-13R α 2 has been explored.^{45–46} This target was specifically chosen for glioblastoma treatment as the receptor is strongly overexpressed in more than 50% of glioblastoma cells, but very mildly expressed in the healthy tissues of the brain. Additionally, previous studies have shown that expression of IL-13R α 2 in glioblastomas correlates with lower survival. In a Phase 1 trial of 65 patients with recurrent glioblastoma, the IL-13R α 2-targeted CAR-T cell therapy showed a median OS of 10.2 months for patients.⁴⁵ A major adverse effect was grade 3 encephalopathy and grade 3 ataxia.⁴⁵ Unfortunately, the higher efficacy of these CAR-T cells depended on a less desired route of administration: the T cells were significantly more effective when delivered through invasive intracranial methods than standard systemic methods.

A bivalent CAR-T therapy was then studied in a Phase 1 clinical trial for glioblastoma.⁴⁷ The bivalent CAR targets IL-13R α 2 and EGFR, using an epitope overexpressed in over 50% of glioblastomas. The immunosuppressive nature of the TME of brain malignancies and differences in antigens between tumors would make this treatment less feasible than most similarly constructed therapies, but the trial still found success with immediate tumor size reductions in all 6 patients. Despite this success, low to moderate neurotoxicity necessitating dexamethasone was observed, and no patients experienced what would qualify as an objective response.

IV.4. EGFRvIII CAR-T cells

The first-in-human trial of the CARv3-TEAM-E T cell therapy was the INCIPIENT trial, and results were recently reported for patients with recurrent glioblastoma.⁴⁸ This CAR-T/BiTE combination therapy targets both the truncated variant of EGFR known as EGFRvIII (expressed on 30% of glioblastoma) and wild-type EGFR which is expressed on glioblastoma. Such targeting occurs through secretion of a T cell-engaging antibody molecule (TEAM) by the engineered cell product. Within days after intraventricular infusion, radiographic reductions in tumor volumes were noted.

Abbott et al. recently discussed another approach for treating EGFRvIII-expressing glioblastoma with T cell-based therapy.⁴⁴ By homing in on the GCT02-binding epitope through traditional CAR-T cell engineering methods, this EGFRvIII CAR-T therapy was shown to attack only the mutant EGFR cells located in the resistant tumors. GCT02-specific

CAR-T cells have been shown to cure murine models of orthotopic intracranial tumors.⁴⁴ This GCT02 epitope has demonstrated greater specificity and targetability than past CAR-T targets in this space. The treatment offers a promising ideal, especially due to the ability of CAR-T cells to cross the blood-brain barrier.

Combinatorial immunotherapy has also been attempted in glioblastoma. EGFRvIII CAR-T cells have been shown to increase the expression of PD-L1 in the TME.⁴⁷ However, in a Phase 1 trial of EGFRvIII-directed CAR-T cells plus pembrolizumab for glioblastoma, there was no increased benefit with repeated infusions of the CAR-T therapy.⁴⁹

IV.5. GD2-CART01

Neuroblastoma is a malignancy of central nervous system origin and represents a major area of pressing need in the pediatric population, given its aggressive nature and the absence of FDA-approved targeted therapies. Treatment of neuroblastoma with CAR-T cells specific to the disialoganglioside GD2 has been attempted for high-risk neuroblastoma. The Phase 1/2 study examined 27 children with severe neuroblastoma who had already received multiple prior lines of therapy.⁵⁰ The CAR-T cells had both CD28 and 4-1BB co-stimulatory domains, and the CAR was directed against GD2. The cell product contained an inducible caspase-9 as a suicide switch to control toxicity. The built-in caspase-9 safety mechanism was activated in 1 of 27 studied patients. Of these 27 patients, 17 achieved an objective response, and 9 of those responses were complete responses. The 3-year OS was 60% and the 3-year event-free survival was 36%. However, CRS again proved to be a major adverse event in 74% of patients.

V. SCIENTIFIC STRATEGIES FOR EXTENDING THE THERAPEUTIC REACH TOWARD SOLID TUMORS

V.1. Optimizing on-target, off-tumor toxicity

One major hurdle to the clinical development of CAR-T therapies in solid tumors is the on-target, off-tumor toxicity that is relevant to various candidate antigens (Figure 2). These can often be dose-limiting toxicities. For example, a significant challenge in designing HER2-specific CAR-T cells is the adverse effect on epithelial barriers, since HER2 is also expressed on normal epithelial cells and is not a tumor-specific antigen. A similar situation is also relevant for TIL therapy in melanoma, in which normal melanocytes can suffer toxicity from TILs, leading to vitiligo as well as other skin issues.⁵¹

Several solutions have been proposed to help limit on-target, off-tumor toxicity of T cell-based therapies. One solution is to design low-affinity CARs that do not strongly target the antigen of interest. An *in vitro* study indicated that CAR-T cells with a single-chain variable fragment (scFv) with low affinity to HER2 exhibited less on-target, off-tumor effect while maintaining robust anti-tumor activity.⁵² Another solution is the design of CAR constructs against tumor-specific antigens or neoantigens. Specific epitopes within antigens may need to be targeted to ensure the desired effects while sparing normal resident tissue. CAR constructs should demonstrate improved functional specificity and enhanced epitope binding toward cancer antigens.

The field may glean from lessons learned in the setting of glioblastoma, in which the design of a CAR construct against GCT02 (an EGFRvIII-specific antigen-binding domain) has led to improved specificity, with less on-target, off-tumor effects.⁴⁴ GCT02-directed CARs have shown no reactivity against primary human keratinocytes, which are known to have high expression of EGFR. Large-scale screening of mutant cell surface antigens may be of particular benefit to molecular target validation across solid tumors. The discovery of natural tumor antigens and non-self epitopes generated by tumor-specific mutations may help guide rational therapeutic design.

V.2. Designing superior CAR constructs

The manufacturing process of CAR constructs plays a critical role in safety and efficacy of CAR-T cells. To date, five generations of CAR-T cells have been developed, and each successive generation has improved safety and/or efficacy based on optimized engineering of the antigen recognition domain (single chain variable fragment), co-stimulatory domain, transmembrane domain, or intracellular domain. Engineering modifications to CAR constructs can be implemented such that the CARs can more effectively recognize tumor cells while sparing healthy cells.

A few examples of engineering modifications of CARs against solid tumors already exist and these technologies are being built upon. In an effort to reduce CAR-T-related toxicities, mechanisms can be designed within constructs to eliminate highly reactive CAR-T cells. Inducible built-in apoptosis mechanisms, such as incorporation of the caspase-9 gene into CARs, can help cease excess adverse effects from CAR-T therapy.⁵⁰ In an effort to reduce the risk of adverse immunologic attack, base-edited CARs can be designed to inactivate key genes to avoid graft-versus-host disease.⁵³ Excess cytokine secretion, which can lead to life-threatening CRS, can be addressed with switchable platforms may help dampen excess cytokine secretion. Improvements in identification of immunogenic antigens may also be valuable for CAR design. Currently, various solid tumor mutations are poorly immunogenic, such as *BRAF* V600E mutation in melanoma. Studies of more immunogenic mutations may help pave the way for target antigen validation. In an effort to overcome tumor antigen heterogeneity, it may be worth designing dual epitope-targeting CARs, which may bypass the problem of antigen escape.

Manufacturing considerations in the future may benefit from including methods that can ensure that CAR-T cells are not eliminated by endogenous immune cells after infusion. The durability of CAR-T cells *in vivo* is a testament to the curative potential of CAR-T therapies.⁵⁴ Persistence of CAR-T cells is essential for long-lasting remissions, and these successes may be dependent on CAR design. Efforts to combat CAR-T cell exhaustion may involve blockade of inhibitory receptors such as PD-1.⁵⁵ These manufacturing considerations should be accounted for when evaluating long-term sustainability across a diverse set of solid tumors.

V.3. Enhancing CAR-T infiltration into the TME

A variety of immunosuppressive cell subsets and cytokines have been found within the solid TME, and these factors contribute to impaired efficacy of cell-based therapies in solid

tumors. Inadequate trafficking of T cells into the solid TME can lead to impaired efficacy. For example, regulatory T cells (T_{regs}), myeloid-derived suppressor cells (MDSCs), and M2 macrophages are prime examples of cell populations that hinder anti-tumor immune responses.⁵⁶ Optimization of the TME would help improve the efficacy of adoptive T cell therapies in solid cancers, and one strategy may be to use monoclonal antibodies to deplete these immunosuppressive subsets within tumors. Immune desert (“cold”) TMEs are also a barrier to the efficacy of CAR-T cell therapies. The analysis of TMEs after CAR-T cell administration has shown a more exhausted T cell phenotype at relapse, so CAR-T exhaustion can be addressed in this context (as above).⁵⁷

On the contrary, tumors with TMEs harboring high or intermediate levels of T cell infiltrates have improved OS and PFS. The “hot” TME appears to be an important mediator of efficacy of CAR-T therapies. Induction of IFN- γ after CAR-T cell administration appears to be associated with enhanced tumor kill.⁴⁵ The TME may also be the key to target antigen validation. For example, IL-13R α 2 is overexpressed in the glioblastoma TME, which makes it an ideal target.

V.4. Overcoming the blood-brain barrier

Numerous solid tumors have predilection for intracranial metastasis. The brain is considered as an immune privileged site and does not typically harbor immune cells. Small molecules and protein-based therapeutics have a challenging time with traversing the blood-brain barrier, as these molecules can only passively diffuse into the destination tissue. As such, CNS-directed therapies are essential in these situations. However, CAR-T therapies have been shown to actively migrate across the blood-brain barrier, with some variability. Furthermore, once CAR-T cells migrate to the CNS, they are thought to be protected against endogenous immune responses that may otherwise limit the survival of CAR-T cells. In glioblastoma, patients who have high CD3⁺/CD8⁺ cell fraction (presumably cytotoxic T cells) have improved survival.⁴⁴

Further research may want to focus on improvement of the transit of CAR-T cells into the CNS. One solution to this challenge is to equip CAR-T cells with adhesion molecules that allows them to cross the blood-brain barrier.^{58–59} This is based on the idea that memory T cells and activated T cells of the CNS harbor adhesion molecules that allow for CNS surveillance.⁵⁹ Intrathecal therapy with CAR-T cells has been performed in a clinical trial to overcome the blood-brain barrier issue.⁴⁷ Intratumoral and intraventricular chemotherapy may help facilitate the efficacy of CAR-T cells against tumors. Additional follow-up will be required in the long term, and larger cohorts should be tested to validate such findings.

VI. OPERATIONAL CHALLENGES IN IMPLEMENTATION

Given the relatively high frequency of solid tumors relative to hematologic malignancies, scalability is an important issue for the rollout of CAR-T and BiTE therapies. Scalability involves consideration of funding, distribution, and implementation of T cell-based therapies across a variety of solid cancers. When considering the complex preparatory regimens and unique toxicity profiles of such therapies, there may be immense challenges to providers and patients located outside of major academic cancer centers. Smaller centers or community

sites may especially lack the specialized personnel and expertise needed to manage side effects such as CRS or ICANS. Therefore, prior to an effective rollout of these therapies as any standard of care for solid tumors, the field may benefit from stronger clinical research efforts aimed at understanding and mitigating toxicity, as well as dedicated training of personnel in the academic and community settings.²

Educational outreach will be a key factor in ensuring that oncologists and other care providers possess proper knowledge of the mechanisms of action, toxicity profiles, and even indications for CAR-T BiTE therapies. Training in standard operating procedures will be necessary. Critical care teams will likely need education on CRS and ICANS management in the event of severe toxicities. The additional education of supportive staff members may help ensure seamless delivery of these therapies in the clinical oncology setting.

Finally, costs of T cell-based therapies in solid malignancies may be very high. Costs may extend beyond staffing obligations, as afami-cel is currently the most expensive CAR-related therapy on the market. Moreover, the delivery of such therapies often requires lengthy hospitalization on the part of the patient along with other preparatory and protective care measures, which present further financial obstacles to the widespread use of T cell-based therapies in solid tumor settings. Therefore, it is essential that relevant training and cost-limiting efforts be made in order to better secure a path to T cell-based therapies as a more realistic option in solid tumor clinical settings.

VII. CONCLUSIONS

As we reflect on the past 7 years since the commercialization of cell-based therapeutics, we can learn significant lessons from the deployment of these therapies in the hematologic malignancies space. Over the past few months, mature clinical trial data has been presented for CAR-T therapies and BiTEs in solid tumors, and these modalities of therapy continue to gain traction. Efforts that paved the way for these therapies in the solid tumor space are to be applauded, though much additional work remains to be done to help ensure extension of the therapeutic reach across many subtypes of solid tumors. A one-size-fits-all approach to this challenge is less likely to work; personalized strategies based on unique disease biology are more likely to be successful. Efforts should focus on identification and validation of the targetable antigen landscape; optimization of CAR manufacturing; minimization on-target, off-tumor effects; and manipulating the immune desert (“cold”) TME. Dual-targeting, trivalent, or pooled CARs may also become available in the future based on interesting pre-clinical data.^{60–61} Novel machine learning-based strategies for antigen or antigen combination identification may be useful. Operational challenges should be addressed, including scalability and cost considerations. The solutions to these problems have been demonstrated for CAR-T therapies used in hematologic malignancies, and the translational potential is high across solid tumors in the coming years.

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DECLARATION OF INTERESTS

SAP served on the Myelodysplastic Syndromes advisory board and the Acute Myeloid Leukemia advisory board for Bristol Myers Squibb, served on the Acute Leukemia Advisory Board for Syndax, and served on the Multiple Myeloma advisory board for Pfizer. SAP received travel support from AAMDSIF in 2024. WKG and YX have no disclosures.

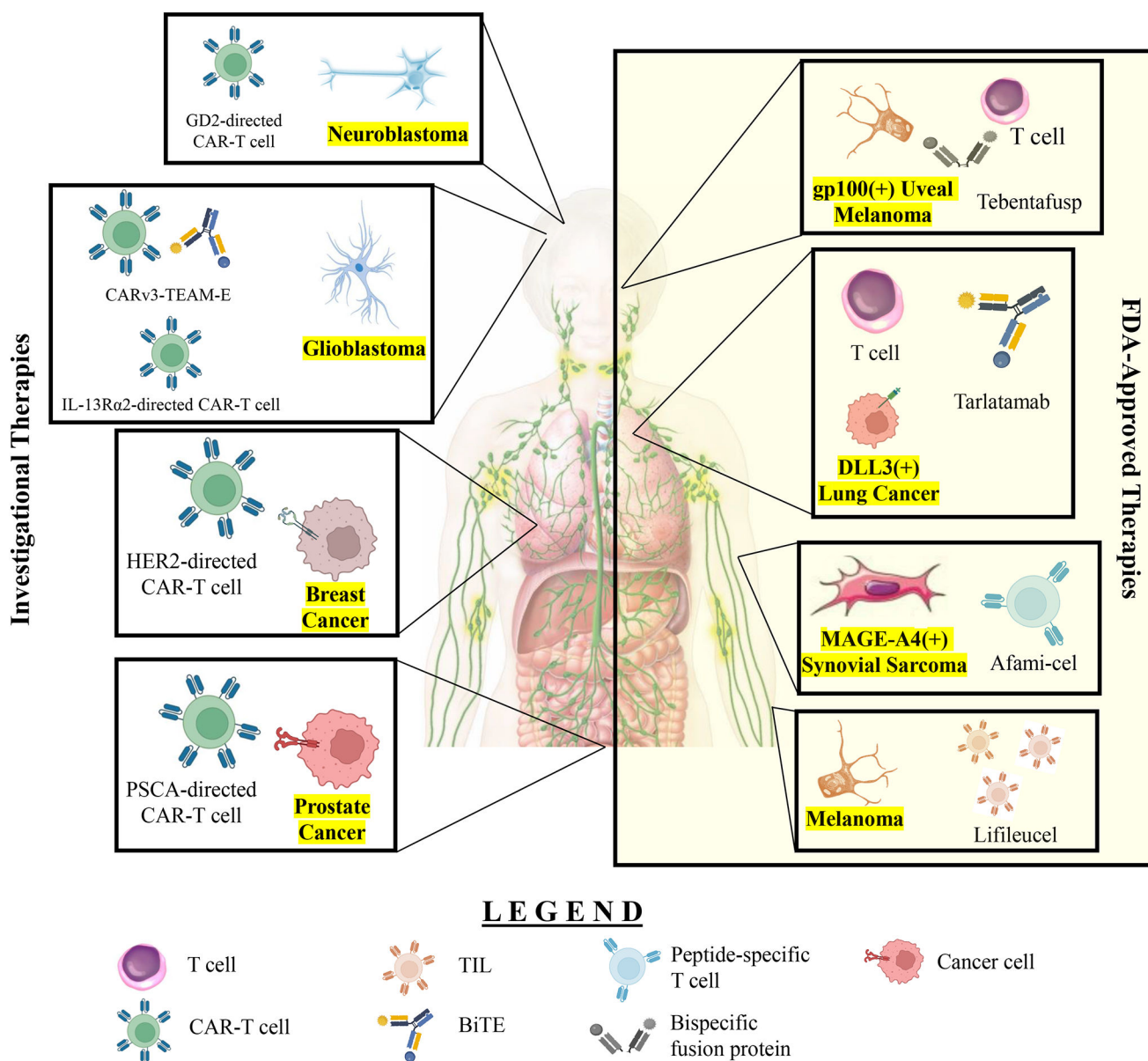
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**FIGURE 1.**

Current landscape of cell-based therapeutics across a variety of solid malignancies. Investigational therapies (*left*) include CAR-T therapies against an array of antigens that are unique to each tissue subtype. FDA-approved therapies are shown (*right*). Diseases are indicated in yellow highlighting. **Abbreviations:** **DLL3**, delta-like ligand 3; **GD2**, disialoganglioside; **HER2**, human epidermal growth factor receptor 2; **MAGE**, melanoma-associated antigen; **PSCA**, prostate stem cell antigen; **TEAM**, T cell-engaging antibody molecule; **TIL**, tumor-infiltrating lymphocyte

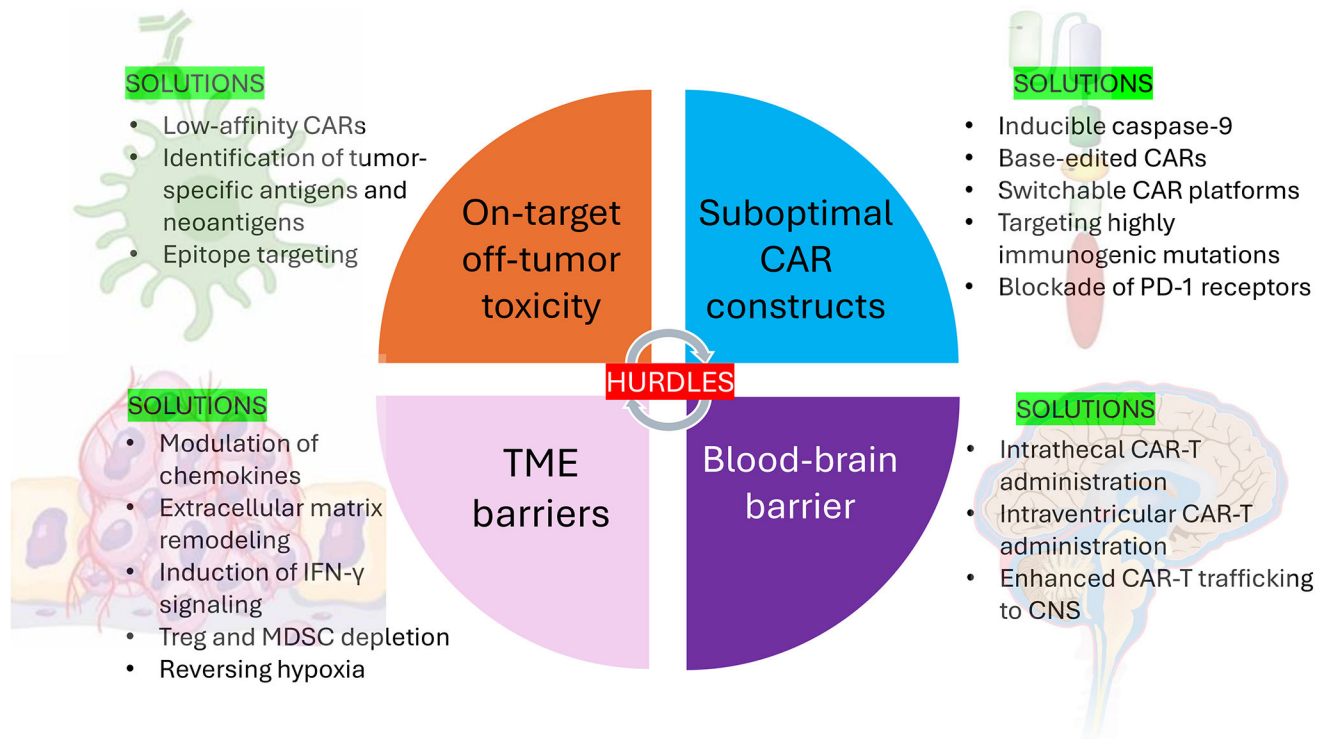


FIGURE 2.

Hurdles and solutions to expanding the therapeutic reach of T cell-based therapies to solid malignancies. Hurdles are shown in the 4 quadrants of the inner circle. Strategies to address these hurdles are shown in the periphery. **Abbreviations:** CNS, central nervous system; **TME**, tumor microenvironment

Table 1:
Current FDA Approvals of CAR-T Cell Therapies and BiTEs for Hematologic Malignancies

Agent	Mechanism of Action	Landmark Trial	Hematologic Disease Indication	Line of Therapy	Reference
CAR-T Therapies					
Tisa-cel	CD19 binding and T cell activation through the 4-1BB and CD3 ζ domains	JULIET, ELIANA, ELARA	LBCL, B cell ALL, FL	LBCL: Third and beyond B cell ALL: Refractory or second and beyond FL: Third and beyond	Schuster <i>et al.</i> , <i>NEJM</i> 2019; Maude <i>et al.</i> , <i>NEJM</i> 2017; Fowler <i>et al.</i> , <i>Nat Med</i> 2021
Axi-cel	CD19 binding and T cell activation through the CD28 and CD3 ζ domains	ZUMA-1, ZUMA-5	LBCL, FL	LBCL: Second and beyond or relapse within 12 months FL: Third and beyond	Neelapu <i>et al.</i> , <i>NEJM</i> 2017; Neelapu <i>et al.</i> , <i>Blood</i> 2023; Jacobsen <i>et al.</i> , <i>Lancet Oncol</i> 2022
Liso-cel	CD19 binding and T cell activation through the 4-1BB and CD3 ζ domains	TRANSCEND NHL 001, TRANSFORM, TRANSCEND FL, TRANSCEND MCL, TRANSCEND CLL 004	LBCL, FL, MCL, CLL	Second or relapse within 12 months	Abramson <i>et al.</i> , <i>Lancet</i> 2020; Kamdar <i>et al.</i> , <i>Lancet</i> 2022; Morschhauser <i>et al.</i> , <i>Nat Med</i> 2024; Wang <i>et al.</i> , <i>JCO</i> 2024; Siddiqi <i>et al.</i> , <i>Lancet</i> 2023
Brexu-cel	CD19 binding and T cell activation through the CD28 and CD3 ζ domains	ZUMA-2, ZUMA-3	B cell ALL, MCL	B cell ALL: Relapsed or refractory MCL: Relapsed or refractory	Wang <i>et al.</i> , <i>NEJM</i> 2020; Shah <i>et al.</i> , <i>Lancet</i> 2021
Ide-cel	BCMA binding and T cell activation through the CD3 ζ domain	KarMMa-1, KarMMa-3	Multiple myeloma	Third and beyond	Munshi <i>et al.</i> , <i>NEJM</i> 2021; Hansen <i>et al.</i> , <i>JCO</i> 2023; Rodriguez-Otero <i>et al.</i> , <i>NEJM</i> 2023
Cilta-cel	BCMA binding and T cell activation through the CD3 ζ domain	CARTITUDE-1, CARTITUDE-4	Multiple myeloma	Second and beyond	Berdeja <i>et al.</i> , <i>Lancet</i> 2021; San-Miguel <i>et al.</i> , <i>NEJM</i> 2023
Obe-cel	CD19 binding and T cell activation through the 4-1BB and CD3 ζ domains	FELIX	B cell ALL	Refractory or second and beyond	Robbie <i>et al.</i> , <i>NEJM</i> 2024
BiTE Therapies					
Blinatumomab	CD3 and CD19 linkage	TOWER	B cell ALL	Relapsed/refractory	Kantarjian <i>et al.</i> , <i>NEJM</i> 2017
Epcoritamab	CD3 and CD20 linkage	EPCORE NHL-1	LBCL, FL	Third and beyond	Thieblemont <i>et al.</i> , <i>JCO</i> 2022; Linton <i>et al.</i> , <i>Lancet Haematol</i> 2024
Glofitamab	CD20 (bivalency) and CD3 (monovalency) linkage	NCT03075696	LBCL	Third and beyond	Dickinson <i>et al.</i> , <i>NEJM</i> 2022
Mosunetuzumab	CD3 and CD20 linkage	NCT02500407	FL	Third and beyond	Budde <i>et al.</i> , <i>Lancet Oncol</i> 2022
Elranatamab	CD3 and BCMA linkage	MagnetisMM-3	Multiple myeloma	Fifth and beyond	Lesokhin <i>et al.</i> , <i>Nat Med</i> 2023; Lesokhin <i>et al.</i> , <i>Blood</i> 2021
Talquetamab	CD3 and GPRC5D linkage	MonumenTAL-1	Multiple myeloma	Fifth and beyond	Krishnan <i>et al.</i> , <i>Blood</i> 2021; Chari <i>et al.</i> , <i>NEJM</i> 2022

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Agent	Mechanism of Action	Landmark Trial	Hematologic Disease Indication	Line of Therapy	Reference
Teclistamab	CD3 and BCMA linkage	MajesTEC-1	Multiple myeloma	Fifth and beyond	Usmani <i>et al.</i> , <i>Lancet</i> 2021; Moreau <i>et al.</i> , <i>NEJM</i> 2022

Abbreviations: **ALL**, acute lymphoblastic leukemia; **BCMA**, B cell maturation antigen; **CLL**, chronic lymphocytic leukemia; **FL**, follicular lymphoma; **LBCL**, large B cell lymphoma; **MCL**, mantle cell lymphoma

Table 2: Current FDA-Approved and Investigational CAR-T Cell Therapies and BiTEs for Solid Tumors as of January 2025

Agent	Mechanism of Action	Landmark Trial	Status	Disease Indication	Line of Therapy	Reference
CAR-T Therapies						
Afami-cel	HLA-restricted T cell therapy with MAGE-A4 affinity	SPEARHEAD-1	FDA-approved	Synovial sarcoma, head and neck cancer, ovarian cancer	Relapsed or refractory	Hong <i>et al.</i> , <i>Nat Med.</i> 2023 D'Angelo <i>et al.</i> , <i>Lancet</i> 2024
Lifileucel	TIL therapy	C-144-01	FDA-approved	Melanoma	Metastatic or unresectable, second and beyond	Sarnaik <i>et al.</i> , <i>J Clin Oncol.</i> 2021
HER2-directed CAR-T	HER2 binding with T cell activation	N/A	Investigational	All HER2-positive cancers	Second and beyond	Oh <i>et al.</i> , <i>Nat Rev Clin Oncol.</i> 2020 Song <i>et al.</i> , <i>Protein Cell.</i> 2018 Xu <i>et al.</i> , <i>Cell Death Dis.</i> 2021
PSCA-directed CAR-T	PSCA binding with T cell activation	N/A	Investigational	Prostate cancer	Metastatic and castration-resistant	Dorff <i>et al.</i> , <i>Nat Med.</i> 2024
EGFR-IL13Rα2 CAR-T	Bivalent targeting of EGFR and IL13Rα2	N/A	Investigational	Glioblastoma	Recurrent	Bagley <i>et al.</i> , <i>Nat Med.</i> 2024
EGFRvIII CAR-T	CAR-T/BiTE combination targeting of GCT02 epitope	INCIPIENT	Investigational	Glioblastoma	Recurrent	Abbott <i>et al.</i> , <i>Clin Transl Immunol.</i> 2023
GD2-CART01	CAR against disialoganglioside GD2	N/A	Investigational	Neuroblastoma	Relapsed or refractory	Del Bufalo <i>et al.</i> , <i>NEJM</i> 2023
BiTEs						
Tarlatacab	CD3 and DLL3 linkage	DeLLphi-301	FDA-approved	Small cell lung cancer	Relapsed or refractory	Ahn <i>et al.</i> , <i>NEJM</i> 2023
Tebentafusp	Anti-CD3 effector conjoined with a soluble affinity-enhanced T cell receptor	IMCgp100-202	FDA-approved	Uveal melanoma	First line	Nathan <i>et al.</i> , <i>NEJM</i> 2021

Abbreviations: CAR, chimeric antigen receptor; **DLL-3**, delta-like ligand 3; **EGFR**, epidermal growth factor receptor; **HER2**, human epidermal growth factor receptor 2; **MAGE**, melanoma-associated antigen; **PSCA**, prostate stem cell antigen; **TIL**, tumor-infiltrating lymphocyte